Yanyan Alexandra Mo

Computational Research Associate
Stanford University

Dissecting gene regulatory mechanisms through functional genomics approaches

ymo@stanford.edu Github

Education

M.S. in Biomedical Engineering Aug. 2023 - May 2025
Johns Hopkins University *Baltimore, MD*

- Dean's Master's Scholarship; Biomedical Engineering Departmental Award

B.S. in Computer Science | Chemical & Biomolecular Engineering Aug. 2020 - May 2023
Johns Hopkins University *Baltimore, MD*

- Graduated with Honors in both degrees

Skills and Interests

Research interests: Gene regulatory network inference, CRISPR-based perturbation screens (Perturb-seq), gene program discovery, agentic AI for biological interpretation
Programming languages: Python, R, C, C++, Java, Bash, HTML
Software engineering: Software and agentic system development, GPU computing, Git, AWS/HPC clusters
Other computer skills: LaTeX, Adobe Illustrator, Microsoft Office

Research Experience

Computational Research Associate July 2025 - Present
Department of Genetics, Stanford University *Stanford, CA*
Advisor: J.Engreitz

- **PerturbNMF:** developed an end-to-end pipeline porting consensus non-negative matrix factorization to GPU with a HALS solver, achieving biological accuracy equivalent to CPU cNMF while reducing compute time up to 20-fold
- **Biological discovery:** applied PerturbNMF to a Perturb-seq dataset of hiPSC-to-endothelial cell differentiation, identifying a novel role for Hippo signaling and charting its effects on cell fate across differentiation time
- **Consortium analysis:** deployed the pipeline across IGVF and MorPhiC datasets, recovering concordant gene programs across single-cell methods; presented results at both consortium meetings
- **Agentic AI systems:** built AGeneTic, an LLM-based agentic framework for automated gene set analysis and literature-grounded program annotation (PaperQA2)

Graduate Research Assistant Oct. 2023 - June 2025
Department of Biomedical Engineering, Johns Hopkins University *Baltimore, MD*
Advisor: P.Cahan

- **CRE-Finder:** developed an information-theoretic algorithm for reconstructing dynamic gene regulatory networks and predicting distal cis-regulatory elements from single-nucleus multi-omics data; first-author manuscript in preparation
- **BMSC multipotency:** performed single-cell RNA-seq analysis of murine bone marrow stromal cells treated with a TEAD1 inhibitor to assess the effect of Hippo signaling on differentiation and multipotency; third author, manuscript under revision

Graduate Research Assistant Mar. 2024 - June 2025
Department of Biomedical Engineering, Johns Hopkins University *Baltimore, MD*
Advisor: W.Grayson

- **ARPA-H proposal:** conceived and drafted sections on predicting craniofacial injury healing from time-series tissue-engineering imaging, 3D microCT, and clinical records
- Conducted feasibility tests with machine learning models predicting bone and vascular healing patterns, and developed domain adaptation strategies for blood vessel segmentation

Undergraduate Research Assistant Jan. 2023 - May 2023
Department of Computer Science, Johns Hopkins University *Baltimore, MD*
Advisor: C.-M.Huang

- **Modeling human reliance on AI:** applied statistical testing and multi-step linear regression for feature selection and modeling to predict human reliance on AI from time-series eye-gaze data; second author, peer-reviewed conference paper

Publications

- **Mo, Y. A.**, Pushkarev, O., Fan, I., Gupta, R., Nguyen, T., King, H., Zeng, T., Everitt, A., Kang, H., Klie, A., Carter, H., Loh, K. M., & Engreitz, J. M., PerturbNMF: an end-to-end pipeline for scalable gene program analysis. ***In preparation*** (2026) [GitHub]
- Cao, S., Ke, S.*, **Mo, Y. A.***, Huang, C., & Liu, A., Eyes are the windows to AI reliance: toward real-time human-AI reliance assessment. ***CHI Conference on Human Factors in Computing Systems*** (2023) [PDF]

Invited presentations

MorPhiC Consortium Face-to-Face Meeting April 2026
- Oral presentation: *Chicago, IL, USA*
  "PerturbNMF: An End-to-End Pipeline for Scalable Gene Program Analysis"

Poster presentations

IGVF Consortium Annual Meeting Sept 2025
Impact of Genomic Variation on Function *Durham, NC, USA*
- Poster: "Performance evaluation of the GPU-accelerated consensus non-negative matrix factorization"

BMES 2024 Oct 2024
Biomedical Engineering Society Annual Meeting *Baltimore, MD, USA*
- Poster: "Discovery of candidate cis-regulatory elements from single-nucleus multi-omics data with CRE-Finder"

Intuitive Surgical Industry Showcase April 2023
Johns Hopkins University *Baltimore, MD, USA*
- Poster: "Fine-tuning BERT for measuring linguistic uncertainty"

Awards, scholarships, and honors

Stanford CVI Travel Award Aug. 2026
Stanford Cardiovascular Institute *Stanford, CA, USA*
- $750 to present PerturbNMF at the IGVF Consortium Annual Meeting, St. Louis, MO

Second Place, QBI Hackathon June 2026
UCSF Quantitative Biosciences Institute *San Francisco, CA, USA*
- $2,000 for AGeneTic, an agentic LLM framework for automated gene set analysis

Dean's Master's Fellowship Sep. 2023 - June 2025
Whiting School of Engineering, Johns Hopkins University *Baltimore, MD, USA*
- $32,000 per year for excellence in academic performance

Biomedical Engineering Departmental Award Sep. 2024
Department of Biomedical Engineering, Johns Hopkins University *Baltimore, MD, USA*
- Awarded for excellence in research

Best Project Award May 2023
Intuitive Surgical Industry Showcase, Johns Hopkins University *Baltimore, MD, USA*
- $200 for the poster "Fine-tuning BERT for measuring linguistic uncertainty"

Center for Student Success Grant Mar. 2022
Life Design Lab, Johns Hopkins University *Baltimore, MD, USA*
- $1,000 supporting a 2022 summer research experience

Teaching Experience

Computational Teaching Assistant Sept 2023 – May 2025
Artificial Intelligence; Intermediate Programming *Computer Science, Johns Hopkins University, Baltimore, MD*